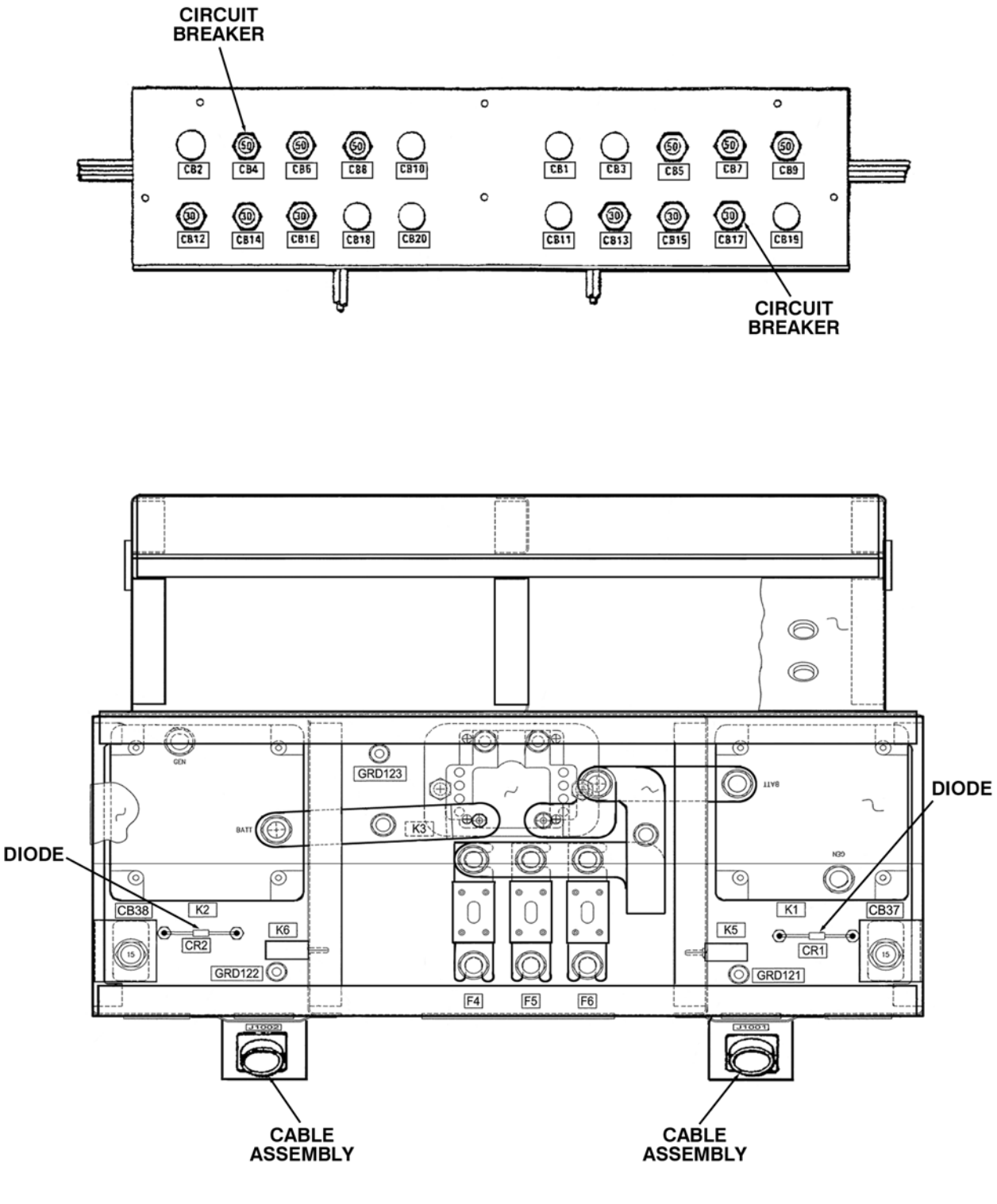
Power Distribution and Generator Control Box (Mod 6/2028) – General Data

1. Description

The power and distribution and generator control box (refer to Figure 1), located in the cabin roof on the right side between stations 162.35 and 180.75, consists of a rectangular aluminum alloy box and cover. The cover is attached to the box by eight quick-release fasteners. Installed in the box are the left and right generator reverse current relays (K1 and K2), the bus tie relay (K3), two generator fail relays (K5 and K6), four blocking diodes (CR1, CR2, CR3 and CR4), a voltage spike suppression diode and the associated electrical wiring. The aircraft electrical power system wiring is routed into the box by two electrical connectors (J1001 and J1002) and through two holes fitted with grommets. A circuit breaker panel mounted on the outside of the box contains six 50-ampere and six 30-ampere bus feed circuit breakers.



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Figure 1

2. Inspection/Check

A. General Visual Inspection and Terminal Torque

Remove the control box cover assembly and inspect the inside components for:

1) General cleanliness. Contamination by metal shavings, dust or fluids. Remove accumulated contamination.

2) Mechanical security.

3) Bus bars and terminal lugs for signs of overheat, burning or arcing.

4) RCR and adjacent components for signs of overheat.

5) Deterioration of rubber grommets and extruded rubber pad.

6) Wiring for:

- General condition
- Routing and security
- Sagging
- Chafing
- Signs of overheat, burning or arcing, discoloration of insulation, deterioration of insulation
- Lacing tape or cable ties missing or incorrectly installed
- Deterioration of previous repairs (e.g. splices)
- Inappropriate repairs

7) Wire terminations and bus bars for:

- Security of terminations, crimped connections and terminal lugs.
- Torque: with aircraft power removed, verify that the nuts on the large RCR and Bus Tie Relay contact posts are torqued within specification (refer to 24-00-00, Electrical Power – General Data, Table 1).

8) Function test DC Generation System (refer to 24-30-00, DC Generation – General Data, Para 5.B.).

B. Deleted

3. Approved Repairs

WARNING - ENSURE ADHESIVE AND SOLVENTS ARE KEPT CLEAR OF FIRE AND OTHER SOURCES OF IGNITION.
- ENSURE SUFFICIENT VENTILATION IS PROVIDED WHEN USING SOLVENTS AND ADHESIVE IN CONFINED AREAS.
- ALWAYS WEAR PROTECTIVE CLOTHING AND GLOVES TO AVOID SKIN CONTACT WITH SOLVENTS AND ADHESIVE.

A. Replace Extruded Rubber Pad

Materials and Equipment required:

- Bostik 1142 (B. B. Chemical Co.)
- Toluol (or Methyl Ethyl Ketone) cleaning solution
- Napthol spirits
- 180 grit paper
- Suitable scraper
- Small brush
- Clean lint free cloths

1) Using a suitable scraper and Toluol cleaning solution, remove all traces of old rubber pad and adhesive from box structure.

2) Lightly scuff surface of new rubber pad with 180 grit paper and clean scuffed area with Napthol spirits and wipe dry with a clean cloth before the solvent evaporates.

3) Clean bond area on box structure with Napthol spirits and wipe dry before solvent evaporates.

   Note After cleaning bond areas, the areas must not be touched with bare hands. If handling is necessary, wear clean cotton gloves.

4) Spread an even coat of Bostik 1142 adhesive to surface of bond areas.

   Note Two coats of adhesive may be required for the rubber pad.

5) Allow the adhesive to dry (i.e. when lightly touched, adhesive will not come away on finger).

6) Mate the rubber pad to the box structure and press down with fingers to ensure overall contact.

7) Allow joint to dry for 2 – 4 hours.

8) Remove excess adhesive with Toluol.